

Girraween High School

2017 Year 12 Half Yearly Examination

Mathematics (2 unit)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Reading time – 5 minutes
- Working time – 2 hours
- Write using a black pen.
- Board-approved calculators may be used.
- The BOSTES *Reference Sheet* may be used (see separate laminated sheet).
- All diagrams are **NOT TO SCALE**.

- **Total Marks: 100**

Use your separate ANSWER BOOKLET to complete this examination.

Write your NAME and your Mathematics TEACHER'S NAME on the cover of your ANSWER BOOKLET(S).

▪ Section 1 (10 Marks):

- Attempt Questions 1 – 10.
- Select A, B, C or D that best answers the question.
- Answer on the *Multiple Choice Answer Sheet* inside your ANSWER BOOKLET.

▪ Section 2 (90 Marks):

- Attempt Questions 11 – 16.
- Show relevant mathematical reasoning and / or calculations.
- Write on both sides of each sheet of paper in your separate ANSWER BOOKLET.
- Start each new question on a NEW PAGE in your ANSWER BOOKLET.
- If you need more paper, ask for another ANSWER BOOKLET.

SECTION I

10 Marks

- Attempt all of Questions 1 – 10.
- Allow about 15 minutes for this section.
- Use the *Multiple Choice Answer Sheet* inside your ANSWER BOOKLET for Questions 1 – 10.

1. Which of the following is e^{-3} written correct to three significant figures?

- (A) 0.049
- (B) 0.050
- (C) 0.0497
- (D) 0.0498

2. For what values of x is the curve $y = 4x^3 - 3x^2$ concave down?

- (A) $x > \frac{1}{4}$
- (B) $x < \frac{1}{4}$
- (C) $x > \frac{3}{4}$
- (D) $x < 0$

3. The primitive function of $x^{-2} - 2$ is:

- (A) $-\frac{1}{x} - 2x + C$
- (B) $\frac{1}{x} - 2x + C$
- (C) $-\frac{1}{3x^3} - 2x + C$
- (D) $\frac{1}{3x^3} - 2x + C$

4. For what values of m will the geometric series $1 + 2m + 4m^2 + 8m^3 + \dots$ have a limiting sum?

(A) $-1 < m < -1$

(B) $-\frac{1}{2} \leq m \leq \frac{1}{2}$

(C) $-\frac{1}{2} < m < \frac{1}{2}$

(D) $m < \frac{1}{2}$

5. What are the coordinates of the focus for the parabola $(x - 2)^2 = -2(y - 3)$?

(A) $\left(-2, 2\frac{1}{2}\right)$

(B) $\left(2, 2\frac{1}{2}\right)$

(C) $(2, 3)$

(D) $(-2, -3)$

6. A bag contains 4 blue marbles and 6 yellow marbles.

Three marbles are selected at random without replacement.

What is the probability that at least one of the marbles selected is blue?

(A) $\frac{1}{6}$

(B) $\frac{1}{2}$

(C) $\frac{5}{6}$

(D) $\frac{29}{30}$

7. Using the trapezoidal rule with 4 subintervals, which expression gives the approximate area under the curve $y = xe^x$ between $x = 1$ and $x = 3$?

- (A) $\frac{1}{4} (e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$
- (B) $\frac{1}{4} (e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$
- (C) $\frac{1}{2} (e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$
- (D) $\frac{1}{2} (e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$

8. Let $a = e^x$.

Which expression is equal to $\log_e(a^2)$?

- (A) e^{x^2}
- (B) e^{2x}
- (C) x^2
- (D) $2x$

9. What is the solution of $5^x = 4$?

- (A) $x = \frac{4}{\log_e(5)}$
- (B) $x = \frac{\log_e(4)}{5}$
- (C) $x = \log_e\left(\frac{4}{5}\right)$
- (D) $x = \frac{\log_e(4)}{\log_e(5)}$

10. What is the value of $\int_{-1}^6 |x - 2| dx$?

(A) $\frac{7}{2}$

(B) $\frac{25}{2}$

(C) $\frac{37}{2}$

(D) $\frac{63}{2}$

The examination continues on the next page.

SECTION II**90 Marks****Attempt all of Questions 11 – 16****Allow about 1 hour and 45 minutes for this section.**

Use your separate ANSWER BOOKLET to complete this Section.

- Show relevant mathematical reasoning and / or calculations.
- Write on both sides of each sheet of paper.
- Start each new question on a NEW PAGE.
- If you need more paper, ask for another ANSWER BOOKLET.

Question 11 (16 Marks)**Marks**

Start on a NEW PAGE in your ANSWER BOOKLET.

(a) Differentiate:

(i) $y = 3x^5 - 2\sqrt{x}$	2
----------------------------	---

(ii) $y = \sqrt{e^{3x}}$	2
--------------------------	---

(iii) $y = \frac{e^x}{e^x + 1}$	3
---------------------------------	---

(b) Find the equation of the tangent to the curve $y = 4e^{3x+1}$ at the y-intercept.	3
---	---

Give your answer in gradient intercept form.

(c) Express $3\log_e 2 + \log_e 3 + \log_e 5 - \log_e 30$ as a single logarithm.	3
--	---

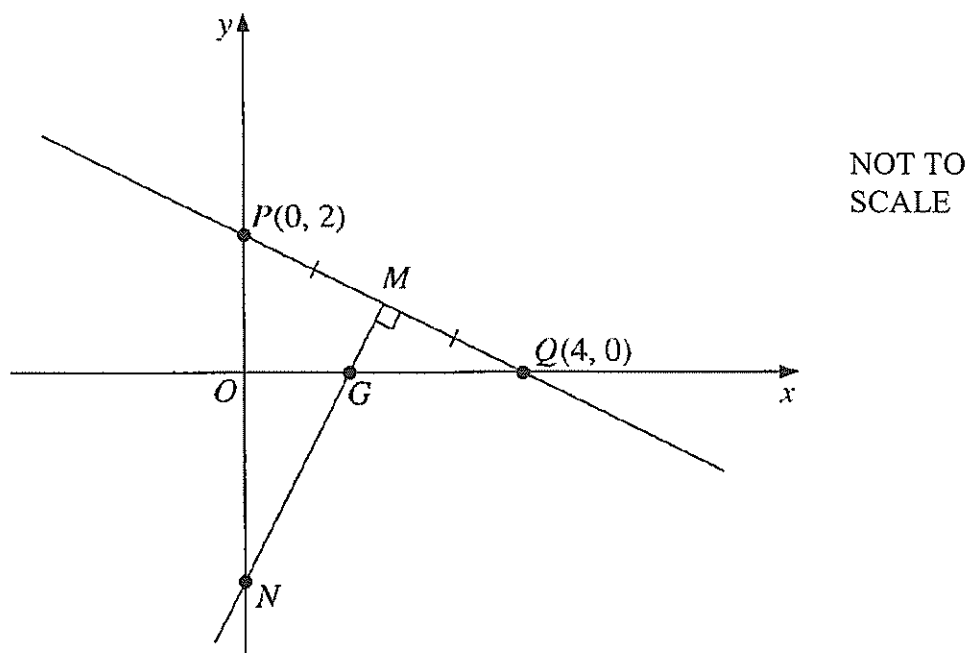
(d) Use Simpson's Rule with five function values to evaluate $\int_1^5 \log_{10} x \, dx$ correct to 4 significant figures.	3
---	---

The examination continues on the next page.

Question 12 (15 Marks)**Marks**

Start on a NEW PAGE in your ANSWER BOOKLET.

(a)



The diagram shows the points $P(0, 2)$ and $Q(4, 0)$.

The point M is the midpoint of PQ .

The line MN is perpendicular to PQ and meets the x -axis at G and the y -axis at N .

- | | | |
|-------|--|---|
| (i) | Show that the gradient of PQ is $-\frac{1}{2}$. | 1 |
| (ii) | Find the coordinates of M . | 2 |
| (iii) | Show that the equation of line MN is $2x - y - 3 = 0$. | 2 |
| (iv) | Show that N has coordinates $(0, -3)$. | 1 |
| (v) | Find the distance NQ . | 2 |
| (vi) | Find the equation of the circle with centre N and radius NQ . | 2 |
| (vii) | Hence show that the circle in part (vi) passes through the point P . | 1 |
- (b) The third term of an arithmetic series is 32 and the sixth term is 17.
- | | | |
|------|--------------------------------------|---|
| (i) | Find the common difference. | 2 |
| (ii) | Find the sum of the first ten terms. | 2 |

The examination continues on the next page.

Question 13 (15 Marks)**Marks**

Start on a NEW PAGE in your ANSWER BOOKLET.

- (a) Find the primitive of $\frac{e^{2x} - 1}{e^x}$. 3
- (b) Find the exact area under the curve $y = \frac{1}{2} (e^x + e^{-x})$ from $x = -2$ to $x = 2$. 3
- (c) The first term of a geometric series is 16 and the fourth term is $\frac{1}{4}$.
- (i) Find the common ratio. 2
- (ii) Find the limiting sum of the series. 2
- (d) A layer of plastic cuts out 15% of the light and lets through the remaining 85%.
- (i) Show that two layers of the plastic let through 72.25% of the light. 2
- (ii) How many layers of the plastic are required to cut out at least 90% of the light? 3

The examination continues on the next page.

Question 14 (15 Marks)**Marks**

Start on a NEW PAGE in your ANSWER BOOKLET.

- (a) Consider the function $y = 4x^3 - x^4$.
- (i) Find the stationary points and determine their nature. 4
- (ii) Sketch the graph of the function, clearly showing the stationary points and the x - and y - intercepts 2
- (b) (i) Differentiate $y = x e^x - e^x$. 2
- (ii) Hence find the exact value of $\int_0^2 x e^x dx$. 2

Question 14 continues on the next page.

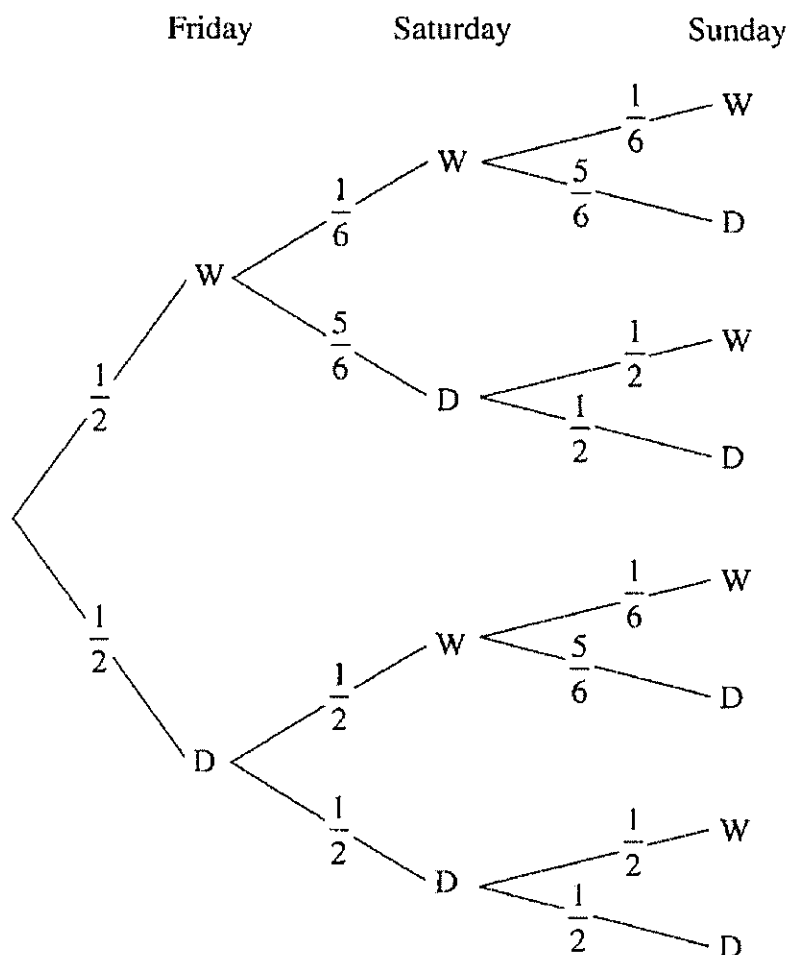
Question 14 (continued)

(c) Weather records for the town of Girraween suggest that:

- if a particular day is wet (W), the probability of the next day being dry is $\frac{5}{6}$
- if a particular day is dry (D), the probability of the next day being dry is $\frac{1}{2}$.

In a specific week Thursday is dry.

The tree diagram shows the possible outcomes for the next three days: Friday, Saturday and Sunday.



- | | | |
|-------|---|---|
| (i) | Show that the probability of Saturday being dry is $\frac{2}{3}$. | 1 |
| (ii) | What is the probability of both Saturday and Sunday being wet? | 2 |
| (iii) | What is the probability of at least one of Saturday and Sunday being dry? | 2 |

End of Question 14.

The examination continues on the next page.

Question 15 (15 Marks)**Marks**

Start on a NEW PAGE in your ANSWER BOOKLET.

- (a) Consider the equation
- $x^2 + (k + 2)x + 4 = 0$
- .

For what values of k does the equation have:

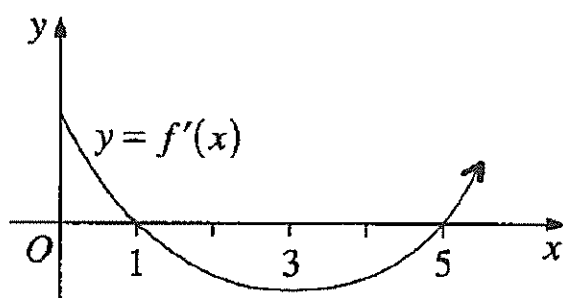
(i) equal roots?

3

(ii) distinct real roots?

2

(b)

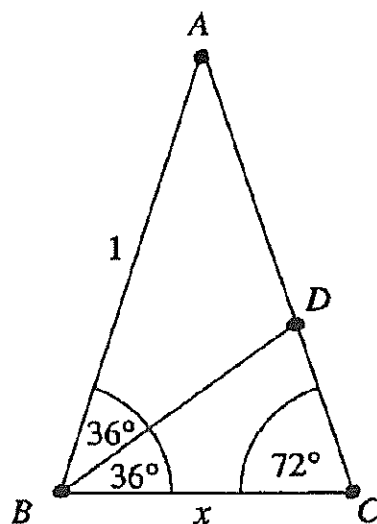
The diagram above shows the graph of the gradient function of the curve $y = f(x)$.

3

For what values of x does $f(x)$ have a local minimum?

Justify your answer.

(c)

NOT TO
SCALEIn the diagram, $\triangle ABC$ is an isosceles triangle where $\angle ABC = \angle BCA = 72^\circ$ and $AB = AC = 1$ unit. $\angle ABC$ is bisected by BD , and $BC = x$ units.

(i) Copy the diagram into your ANSWER BOOKLET.

1

(ii) Show that triangles $\triangle ABC$ and $\triangle BCD$ are similar.

3

(iii) By using (ii), find the exact value of x .

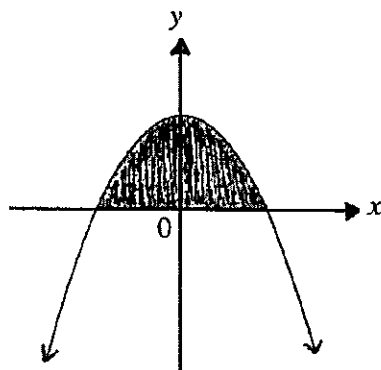
3

The examination continues on the next page.

Question 16 (14 Marks)**Marks**

Start on a NEW PAGE in your ANSWER BOOKLET.

(a)

NOT TO
SCALE

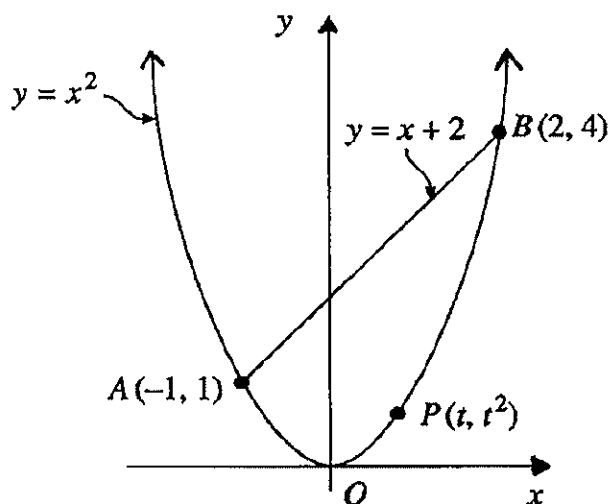
The shaded region lying between the curve $y = 1 - x^2$ and the x -axis is rotated about the x -axis.

4

Find the exact volume of the solid formed.

(b) Solve $2 \log_e x = \log_e (x + 12)$ **3**

(c)

NOT TO
SCALE

In the diagram, $A(-1, 1)$ and $B(2, 4)$ are the points of intersection of the parabola $y = x^2$ with the line $y = x + 2$.

The point $P(t, t^2)$ is a variable point on the parabola below the line.

- (i) Find the area of the parabolic segment APB , that is, the area below the line and above the parabola. **3**
- (ii) Show that the maximum area of the triangle $\triangle APB$ is three-quarters of the area of the parabolic segment APB . **4**

End of examination.

Multiple choice:

- ① D ② B ③ A ④ C ⑤ B
⑥ C ⑦ A ⑧ D ⑨ D ⑩ B

Ques 1: $e^{-3} = 0.04978706837$
↑↑↑

$= 0.0498$ (3 sig fig's) (D)

Ques 2: $y = 4x^3 - 3x^2$

$\frac{dy}{dx} = 12x^2 - 6x$

$\frac{d^2y}{dx^2} = 24x - 6$

for concave down, $\frac{d^2y}{dx^2} < 0$

i.e. $24x - 6 < 0$

$24x < 6$

$x < \frac{1}{4}$

(B)

Ques 3: $\int x^{-2} - 2 dx = \frac{x^{-1}}{-1} - 2x + C$
 $= -\frac{1}{x} - 2x + C$ (A)

Ques 4: $1 + 2m + 4m^2 + 8m^3 + \dots$

Geometric series with $a=1, r=2m$

for limiting sum, $|r| < 1$

i.e. $-1 < 2m < 1$

$-\frac{1}{2} < m < \frac{1}{2}$

(C)

Ques 5: $(x-2)^2 = -2(y-3)$

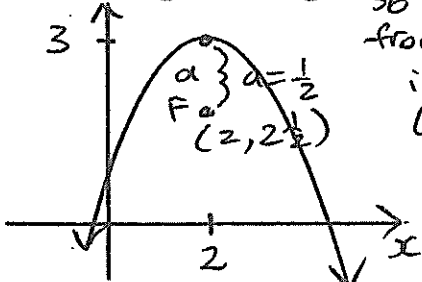
this is in the form $(x-x_1)^2 = -4a(y-y_1)$

which is concave down parabola, vertex (x_1, y_1)

so $4a = 2$ so concave down, vertex $(2, 3)$
 $a = \frac{1}{2}$

So focal length $\Rightarrow a = \frac{1}{2}$

So focus from sketch is $(2, 2\frac{1}{2})$



Ques 6: $P(\text{at least 1 blue})$

$= 1 - P(\text{No blue, i.e. all 3 yellow})$

$= 1 - \left(\frac{6}{10} \times \frac{5}{9} \times \frac{4}{8}\right)$

$= 1 - \frac{1}{6}$

$= \frac{5}{6}$

(C)

Ques 7: $n = \frac{b-a}{n} = \frac{3-1}{4} = \frac{1}{2} = 0.5$

$y = xe^x$ (4 subintervals = 2 applications)

x	1	1.5	2	2.5	3
f(x)	$1e^1$	$1.5e^{1.5}$	$2e^2$	$2.5e^{2.5}$	$3e^3$

Area $\approx \frac{0.5}{2} [f(1) + 2f(1.5) + 2f(2) + 2f(2.5) + f(3)]$

$\approx \frac{1}{4} [1e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3]$ (A)

Ques 8: given $a = e^x$

so $\log_e(a^2) = 2\log_e(a)$

$= 2\log_e e^x$

$= 2x \cdot \log_e e$

$= 2x \times 1$

$= 2x$

(D)

Ques 9: $5^x = 4$

Taking logarithms to base e both sides:

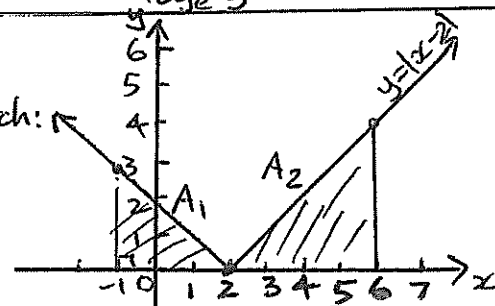
$\log_e 5^x = \log_e 4$

$x \log_e 5 = \log_e 4$

$\therefore x = \frac{\log_e 4}{\log_e 5}$

Ques 10:

$y = |x-2|$ is shown in sketch:



$\therefore \int_{-1}^6 |x-2| dx = \text{Area of two shaded triangles}$
 $= A_1 + A_2$
 $= \frac{1}{2} \times 3 \times 3 + \frac{1}{2} \times 4 \times 4$
 $= \frac{9}{2} + \frac{16}{2}$
 $= \frac{25}{2}$ (B)

QUESTION 11 (16 Marks)

(a)(i) $y = 3x^5 - 2\sqrt{x}$

ie. $y = 3x^5 - 2x^{\frac{1}{2}}$

$$\begin{aligned}\therefore \frac{dy}{dx} &= 15x^4 - 2 \cdot \frac{1}{2} x^{-\frac{1}{2}} \\ &= 15x^4 - x^{-\frac{1}{2}} \\ &= 15x^4 - \frac{1}{\sqrt{x}} \quad (2m)\end{aligned}$$

(ii) $y = \sqrt{e^{3x}}$

$y = (e^{3x})^{\frac{1}{2}}$

ie. $y = e^{\frac{3x}{2}}$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{3}{2} e^{\frac{3x}{2}} \\ &= \frac{3}{2} \sqrt{e^{3x}} \quad (2m)\end{aligned}$$

(iii) $y = \frac{e^x}{e^x + 1}$

Let $u = e^x$ $v = e^x + 1$

$\frac{du}{dx} = e^x$ $\frac{dv}{dx} = e^x$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \\ &= \frac{(e^x + 1)(e^x) - (e^x)(e^x)}{(e^x + 1)^2} \\ &= \frac{e^{2x} + e^x - e^{2x}}{(e^x + 1)^2}\end{aligned}$$

$$\therefore \frac{dy}{dx} = \frac{e^x}{(e^x + 1)^2} \quad (3m)$$

(b) $y = 4e^{3x+1}$

$\frac{dy}{dx} = 4(3)(e^{3x+1})$

$\frac{dy}{dx} = 12(e^{3x+1})$

At y-intercept, $x=0$, $y = 4e^{0+1}$
 $y = 4e$

when $x=0$, gradient tangent $m = 12e^{0+1}$
 $m = 12e$

$(0, 4e)$, $m = 12e$:

Equation tangent: $y - y_1 = m(x - x_1)$
 $y - 4e = 12e(x - 0)$

$\therefore y = 12ex + 4e \quad (3m)$

Ques 11 continued:

$$\begin{aligned}(c) \quad &3 \log_e 2 + \log_e 3 + \log_e 5 - \log_e 30 \\ &= \log_e 2^3 + \log_e 3 + \log_e 5 - \log_e 30 \\ &= \log_e \left(\frac{8 \times 3 \times 5}{30} \right) \\ &= \log_e 4 \quad (3m)\end{aligned}$$

(d) $f(x) = \log_{10} x$

$h = \frac{b-a}{n}$ 5 function values

$= \frac{5-1}{4} = 4$ subintervals

$\therefore h = 1$ (= 2 applications)

For $f(x) = \log_{10} x$, construct a table of values/weightings:

Weighting	①	4	2	4	①
x	1	2	3	4	5
$f(x)$	$\log_{10} 1$	$\log_{10} 2$	$\log_{10} 3$	$\log_{10} 4$	$\log_{10} 5$

The formula for Simpson's Rule using the weighting method is:

$$\int_a^b f(x) dx \approx \frac{h}{3} \left\{ f(a) + 4 \times (\text{function values}) + 2 \times (\text{function values}) + f(b) \right\}$$

Using values from the table:

$$\int_1^5 \log_{10} x dx$$

$$\approx \frac{1}{3} \left\{ \log_{10} 1 + 4 \times (\log_{10} 2 + \log_{10} 4) + 2 \times (\log_{10} 3) + \log_{10} 5 \right\}$$

$= \frac{1}{3} \times 5.265572462$

$= 1.755190821 \dots$

 $\uparrow \uparrow \uparrow$

$= 1.755 \text{ (4 sig figs)} \quad (3m)$

QUESTION 12 (15 Marks)(a) (i) gradient PQ $P(0,2)$

$$m_{PQ} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{0-2}{4-0}$$

$$m_{PQ} = -\frac{2}{4}$$

$$m_{PQ} = -\frac{1}{2} \text{ as required } (1m)$$

(ii) midpoint PQ

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{0+4}{2}, \frac{2+0}{2} \right)$$

 $\therefore M$ has coordinates $(2,1)$ (2m)(iii) $MN \perp PQ$ So if $m_{PQ} = -\frac{1}{2}$ from (i),

$$m_{PQ} = 2 \text{ (since } m_1 m_2 = -1)$$

So $m_{PQ} = 2$, passing through $(2,1)$:

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 2)$$

$$y - 1 = 2x - 4$$

 $\therefore 2x - y - 3 = 0$ is equation of MN (2m)(iv) N is on the y -axis, so

$$\text{when } x=0: 0 - y - 3 = 0$$

$$y = -3$$

 $\therefore N$ has coordinates $(0,-3)$ (1m)

$$\begin{aligned} \text{(v)} \quad d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad N(0,-3) \\ &= \sqrt{(4-0)^2 + (0-(-3))^2} \quad Q(4,0) \\ &= \sqrt{4^2 + 3^2} \\ &= \sqrt{25} \end{aligned}$$

 $\therefore$ distance $NQ = 5$ units (2m)(vi) centre $N(0,-3)$, radius 5 units

$$(x - 0)^2 + (y - (-3))^2 = 5^2$$

$$\therefore x^2 + (y+3)^2 = 25$$

is equation of circle (2m)

Ques 12 continued:

(12)(a)(vii)

Point $P(0,2)$ lies on circle

$$x^2 + (y+3)^2 = 25$$

METHOD 1Distance PN

$$= 2+3$$

$$= 5$$

 $=$ radius of circle $\therefore$ circle passes through P (1m)METHOD 2 $P(0,2)$ Substitute $x=0, y=2$:

$$(0)^2 + (2+3)^2 = 25$$

$$25 = 25$$

$$(12)(b)(i) T_3 = a + 2d = 32 \quad (1)$$

$$T_6 = a + 5d = 17 \quad (2)$$

$$(2) - (1): 3d = -15$$

 $\therefore$ Common difference, $d = -5$ (2m)

$$(ii) S_n = \frac{n}{2} (2a + (n-1)d)$$

Need value of a , so from (1),

$$a + 2 \times (-5) = 32$$

$$a = 42$$

$$\therefore S_{10} = \frac{10}{2} (2 \times 42 + 9 \times (-5))$$

$$= 5 \times (39)$$

$$\therefore S_{10} = 195 \quad (2m)$$

QUESTION 13 (15 Marks)

$$(a) \int \frac{e^{2x} - 1}{e^x} dx$$

$$= \int \frac{e^{2x}}{e^x} - \frac{1}{e^x} dx$$

$$= \int e^x - e^{-x} dx$$

$$= e^x + e^{-x} + C$$

(3m)

Ques 13 continued:

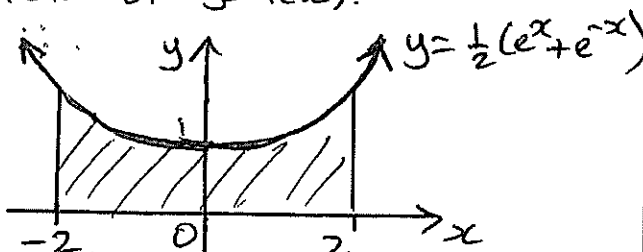
13)(b) $y = \frac{1}{2}(e^x + e^{-x})$
 y-intercept: $x=0, y = \frac{1}{2}(e^0 + e^0)$
 $= \frac{1}{2} \times 2$

$$y = 1$$

Even-function: $f(-x) = \frac{1}{2}(e^{-x} + e^{-(-x)})$
 $= \frac{1}{2}(e^{-x} + e^x)$

$\therefore$ symmetrical about y -axis

Sketch of $y=f(x)$:



METHOD 1:

As $y = \frac{1}{2}(e^x + e^{-x})$ is an even function,

$$\begin{aligned} \text{Area} &= 2 \times \int_0^2 \frac{1}{2}(e^x + e^{-x}) dx \\ &= [e^x - e^{-x}]_0^2 \\ &= (e^2 - e^{-2}) - (e^0 - e^0) \\ &= e^2 - \frac{1}{e^2} - (1 - 1) \end{aligned}$$

$\therefore \text{Area} = e^2 - \frac{1}{e^2}$ square units

METHOD 2:

$$\begin{aligned} \text{Area} &= \int_{-2}^2 \frac{1}{2}(e^x + e^{-x}) dx \\ &= \frac{1}{2} [e^x - e^{-x}]_{-2}^2 \\ &= \frac{1}{2} [(e^2 - e^{-2}) - (e^{-2} - e^2)] \\ &= \frac{1}{2} [e^2 - \frac{1}{e^2} - \frac{1}{e^2} + e^2] \\ &= \frac{1}{2} [2e^2 - \frac{2}{e^2}] \\ &= \frac{1}{2} \times 2(e^2 - \frac{1}{e^2}) \end{aligned}$$

$\text{Area} = e^2 - \frac{1}{e^2}$ square units

Ques 13 continued:

13)(c) $T_1 = a = 16$ (1)

(i) $T_4 = ar^3 = \frac{1}{4}$ (2)

Substitute $a=16$ into (2),

$$16r^3 = \frac{1}{4}$$

$$r^3 = \frac{1}{64}$$

$\therefore r = \frac{1}{4}$ (2m)

(c)(ii) $a = 16, r = \frac{1}{4}$

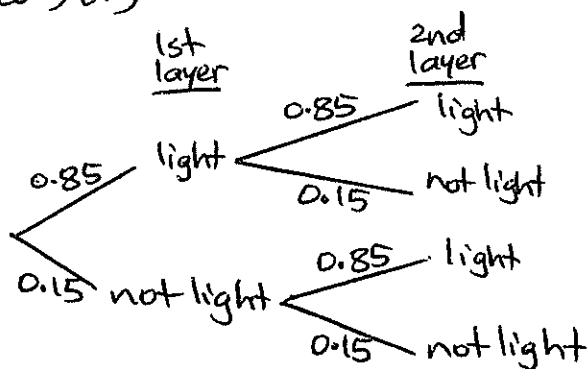
$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{16}{1 - \frac{1}{4}}$$

$$= \frac{16}{\frac{3}{4}}$$

$\therefore$ limiting sum $= 21\frac{1}{3}$ (2m)

(d)(i)



Amount of light for 2 layers $= 0.85 \times 0.85$
 $= 0.7225$
 $= 72.25\%$ (2m)

(ii) Need to find n layers of plastic that allow at least 10% of light:

$$0.85^n \leq 0.1$$

Taking logarithm to base e of both sides:

$$\log_e 0.85^n \leq \log_e 0.1$$

$$n \log_e 0.85 \leq \log_e 0.1$$

Note that $\log_e 0.85$ is negative, so

$$n \geq \frac{\log_e 0.1}{\log_e 0.85}$$

$$n \geq 14.16810399$$

$\therefore$ require at least 15 layers. (3m)

QUESTION 14 (15 Marks)

(a) (i) $y = 4x^3 - x^4$

$$\frac{dy}{dx} = 12x^2 - 4x^3$$

$$\frac{d^2y}{dx^2} = 24x - 12x^2$$

Stationary points when $\frac{dy}{dx} = 0$:

$$12x^2 - 4x^3 = 0$$

$$4x^2(3 - x) = 0$$

$$\therefore x = 0 \text{ or } x = 3$$

when $x = 0$:

$$y = 0$$

when $x = 3$:

$$y = 4x^3 - x^4 = 27$$

 $\therefore$ two stationary points are $(0,0)$ and $(3,27)$

At $x=0$: $\frac{d^2y}{dx^2} = 0$

This is a possible point of inflexion.
check for change of concavity.when $x = -1$:

$$\frac{d^2y}{dx^2} = -24 - 12 = -36$$

$$\frac{d^2y}{dx^2} < 0$$

concave down

when $x = 1$:

$$\frac{d^2y}{dx^2} = 24 - 12 = 12$$

$$\frac{d^2y}{dx^2} > 0$$

concave up

 $\therefore$ change of concavity and $\frac{d^2y}{dx^2} = 0$ when $x = 0$. $\therefore$ At $(0,0)$, horizontal point of inflexion.

At $x=3$: $\frac{d^2y}{dx^2} = 24x - 12x^2 = -36$

$$\therefore \frac{d^2y}{dx^2} < 0$$

 $\therefore$ concave DOWN $\checkmark \checkmark$ $\therefore$ At $(3,27)$, 4m
MAXIMUM turning point.Ques 14(a)(ii) continued:14)(a)(ii) x -intercepts when $y=0$:

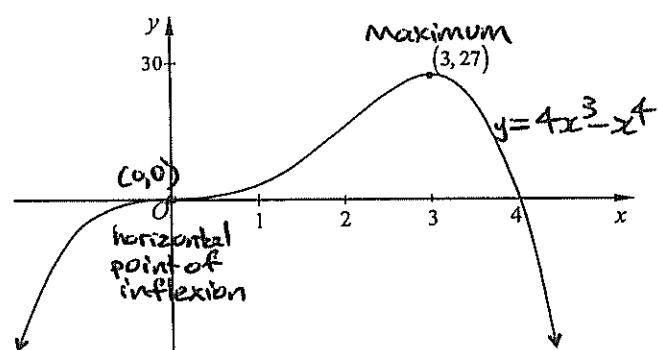
$$4x^3 - x^4 = 0$$

$$x^3(4 - x) = 0$$

so x -intercepts at $x=0$, and $x=4$ y -intercepts when $x=0$

$$y = 0 - 0$$

$$y = 0$$

So y -intercept at $(0,0)$ $(3,27)$ Maximum turning point $(0,0)$ horizontal point of inflexion.So sketch of $y = 4x^3 - x^4$ is:2m

14)(b)(i) $y = xe^x - e^x$

so $\frac{dy}{dx} = \frac{d}{dx}(xe^x) - \frac{d}{dx}(e^x)$

$$= [x \cdot e^x + e^x \cdot 1] - e^x$$

$$= xe^x + e^x - e^x$$

$$\therefore \frac{dy}{dx} = xe^x$$
 2m

(ii) $\therefore \int xe^x dx = xe^x - e^x + C$

$$\text{so } \int_0^2 xe^x dx = [xe^x - e^x]_0^2$$

$$= (2e^2 - e^2) - (0 - e^0)$$

$$= e^2 + 1$$
2m

Ques 14 continued:

$$\begin{aligned}
 (14)(c)(i) P(\text{Sat dry}) &= P(WD) \text{ or } P(DD) \\
 &= \frac{1}{2} \times \frac{5}{6} + \frac{1}{2} \times \frac{1}{2} \\
 &= \frac{5}{12} + \frac{1}{4} \\
 \therefore P(\text{Sat dry}) &= \frac{2}{3} \text{ as required (1m)}
 \end{aligned}$$

$$\begin{aligned}
 (c)(ii) P(\text{Sat} + \text{Sun Wet}) &= P(WWW) \text{ or } P(DWW) \\
 &= \frac{1}{2} \times \frac{1}{6} \times \frac{1}{6} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{6} \\
 &= \frac{1}{72} + \frac{1}{24} \\
 &= \frac{1}{18} \quad (2m)
 \end{aligned}$$

$$\begin{aligned}
 (c)(iii) P(\text{at least one of Sat or Sun DRY}) &= 1 - P(\text{both wet}) \\
 &= 1 - P(\text{Sat} + \text{Sun wet}) \\
 &= 1 - \frac{1}{18} \\
 &= \frac{17}{18} \quad (2m)
 \end{aligned}$$

QUESTION 15 (14 Marks)

$$(a) \quad x^2 + (k+2)x + 4 = 0$$

$$\begin{aligned}
 (i) \quad \Delta &= b^2 - 4ac \\
 &= (k+2)^2 - 4 \times 1 \times 4 \\
 &= k^2 + 4k + 4 - 16
 \end{aligned}$$

$$\therefore \Delta = k^2 + 4k - 12$$

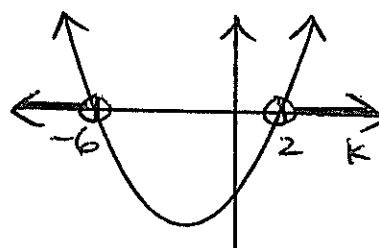
For equal roots, $\Delta = 0$

$$\text{i.e. } k^2 + 4k - 12 = 0$$

$$(k+6)(k-2) = 0 \quad (3m)$$

 $\therefore$ for equal roots, $k = -6$ or $k = 2$ Ques 15 (a) continued:

$$\begin{aligned}
 (15)(a)(ii) \text{ For distinct real roots, } &\Delta > 0 \\
 \text{i.e. } k^2 + 4k - 12 > 0 & \\
 (k+6)(k-2) > 0 &
 \end{aligned}$$



This solution is

$$k < -6 \text{ or } k > 2$$

(2m)

(15)(b) from the diagram, stationary points are when $f'(x) = 0$

which occurs at $x = 1$ and $x = 5$. To find the MINIMUM turning point, use either First or Second derivative test.

First derivative Test (METHOD 1)At $x = 1$:

from graph:

when $x = 0$,

$$f'(0) > 0$$

+ / -

when $x = 2$,

$$f'(2) < 0$$

 $\therefore$ At $x = 1$, MAXIMUM turning point.At $x = 5$:

From graph:

when $x = 4$,

$$f'(4) < 0$$

- / +

when $x > 5$,

$$f'(x) > 0$$

$\therefore$ At $x = 5$, MINIMUM turning point (as required)

(3m)

Ques 15(b) continued:Second Derivative Test (METHOD 2)

At $x=1$: $f'(x)$ is decreasing from graph
 i.e. gradient of $f'(x)$ is negative -

$$\text{i.e. } f''(x) < 0$$

$\therefore f''(x)$ is concave DOWN

$\therefore$ At $x=1$, MAXIMUM turning point.

At $x=5$:

$f'(x)$ is increasing from graph

i.e. gradient of $f'(x)$ is positive -

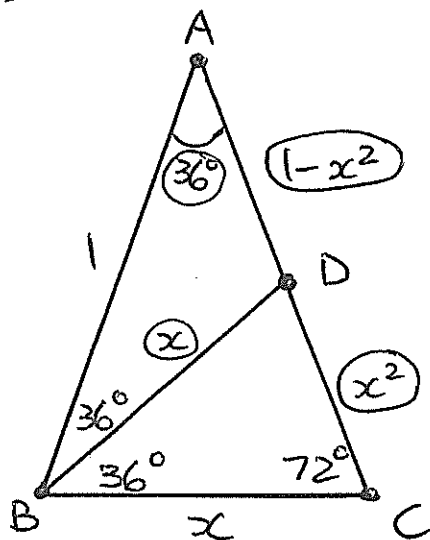
$$\text{i.e. } f''(x) > 0$$

$\therefore f''(x)$ is concave UP.

$\therefore$ at $x=5$, MINIMUM turning point.

15)(c)

(i)



Copy diagram

(1m)

Ques 15(c) continued:

(5)(c)(ii)

In $\triangle ABC$: $\angle ABC = \angle ACB = 72^\circ$ (given)

In $\triangle BCD$:

$$\begin{aligned} \angle BDC &= 180^\circ - (36^\circ + 72^\circ) \\ &= 180^\circ - 108^\circ \text{ (angle sum of triangle)} \\ \therefore \angle BDC &= 72^\circ \end{aligned}$$

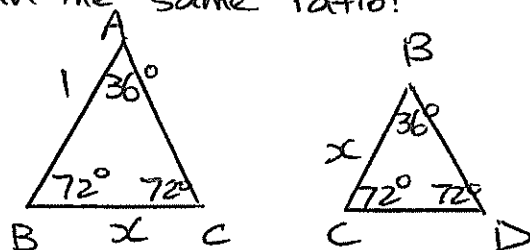
So in $\triangle BCD$: $\angle BDC = \angle BCD = 72^\circ$

$\therefore \triangle ABC \parallel \triangle BCD$ (equiangular)

(3m)

(c)(iii)

Since $\triangle ABC$ and $\triangle BCD$ are similar, the corresponding sides are in the same ratio:



$$\text{So } \frac{BC}{AB} = \frac{x}{1} = \frac{CD}{BC}$$

$$\text{i.e. } \frac{CD}{BC} = \frac{x}{1}$$

$$\begin{aligned} \text{but } BC &= x, \text{ so } \frac{CD}{x} = \frac{x}{1} \\ \therefore CD &= x^2 \end{aligned}$$

Now since $BC = x$, we have $CD = x^2$

But $\triangle ABC$ is isosceles, so if $AB = 1$, $AC = 1$ (sides opposite equal angles are equal)

$$\text{But } AC = AD + CD$$

$$1 = AD + x^2$$

$$\therefore AD = 1 - x^2 \text{ (angle sum of triangle)}$$

$$\text{In } \triangle ABC, \angle BAC = 180^\circ - (72^\circ + 72^\circ) = 36^\circ$$

$\therefore$ in $\triangle ABD$, $\angle ABD = 36^\circ = \angle BAD$

$\therefore \triangle ABD$ is isosceles

$\therefore AD = BD$ (sides opposite equal angles are equal)

Ques 15(c)(iii) continued:In $\triangle BCD$:

$$\angle BDC = 180^\circ - (36^\circ + 72^\circ) = 72^\circ$$

 $\therefore \triangle BCD$ is isosceles $\therefore BC = x = BD$ (sides opposite equal angles are equal)Now since $AD = BD$,we have $AD = 1 - x^2$ and $BD = x$

$$\therefore 1 - x^2 = x$$

$$\text{or } x^2 - x - 1 = 0$$

from quadratic equation,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-1) \pm \sqrt{1 - 4 \times 1 \times (-1)}}{2}$$

$$x = \frac{1 \pm \sqrt{5}}{2}$$

but x is a length, so $x > 0$

so only solution is

$$x = \frac{1 + \sqrt{5}}{2}$$

(3m)

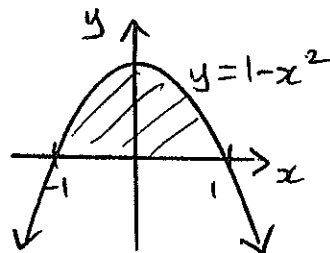
QUESTION 16 (14 marks)

(a) $y = 1 - x^2$

when $y = 0$, $1 - x^2 = 0$

$(1-x)(1+x) = 0$

$x = 1 \text{ or } x = -1$



$$\therefore \text{Volume} = \pi \int_{-1}^1 y^2 dx$$

$$= \pi \int_{-1}^1 (1 - x^2)^2 dx$$

$$= \pi \int_{-1}^1 1 - 2x^2 + x^4 dx$$

$$= \pi \left[x - \frac{2x^3}{3} + \frac{x^5}{5} \right]_{-1}^1$$

$$= \pi \left[\left(1 - \frac{2}{3} + \frac{1}{5}\right) - \left(-1 + \frac{2}{3} - \frac{1}{5}\right) \right]$$

$$= \pi \left[\frac{8}{15} - \left(-\frac{8}{15}\right) \right]$$

$$= \frac{16\pi}{15} \text{ cubic units } (4m)$$

(b) $2 \log_e x = \log_e (x+12)$

from logarithm laws,

$\log_e x^2 = \log_e (x+12)$

Equating logarithms:

$x^2 = x + 12$

$x^2 - x - 12 = 0$

$(x-4)(x+3) = 0$

so $x = -3$ or $x = 4$

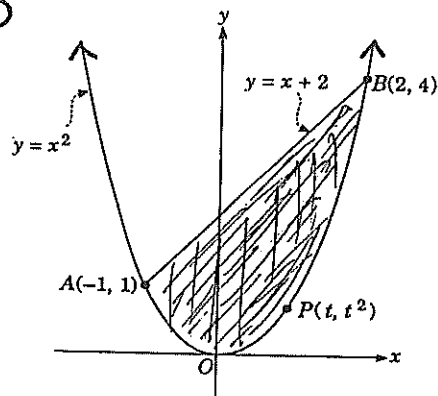
 $x = -3$ is NOT a solution since for logarithms, $x > 0$.

So only solution is $x = 4$ (3m)

Ques 16 continued:

16)cc)

1i)



Area of parabolic segment APB

$$= (\text{Area under line } y=x+2) - (\text{Area under parabola } y=x^2)$$

$$= \int_{-1}^2 (x+2) - x^2 dx$$

$$= \int_{-1}^2 x+2-x^2 dx$$

$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2$$

$$= \left(\frac{4}{2} + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right)$$

$$= \frac{9}{2}$$

$$\therefore \text{Area} = 4\frac{1}{2} \text{ square units } (3m)$$

c) (ii) To find area ΔAPB :

① Need length of AB

② perpendicular distance from $P(t, t^2)$ to line AB ($y=x+2$)

① Length of AB

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(2 - (-1))^2 + (4 - 1)^2}$$

$$= \sqrt{3^2 + 3^2}$$

$$= \sqrt{18}$$

$$= \sqrt{9} \times \sqrt{2}$$

$$\therefore \text{length AB} = 3\sqrt{2} \text{ units}$$

Question 16(c) (ii) continued:

② Perpendicular distance from $P(t, t^2)$ to line AB (ie. $x - y + 2 = 0$)

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$d = \frac{|1 \times t - 1 \times t^2 + 2|}{\sqrt{1^2 + (-1)^2}}$$

$$d = \frac{|t - t^2 + 2|}{\sqrt{2}}$$

$$\therefore d = \frac{t - t^2 + 2}{\sqrt{2}}$$

$$\therefore \text{Area } \Delta APB = \frac{1}{2} \times AB \times d$$

$$= \frac{1}{2} \times 3\sqrt{2} \times \frac{(t - t^2 + 2)}{\sqrt{2}}$$

$$\therefore A = \frac{3}{2} (t - t^2 + 2)$$

For maximum area, need to solve $\frac{dA}{dt} = 0$.

$$\frac{dA}{dt} = \frac{3}{2} (1 - 2t)$$

$$\frac{dA}{dt} = 0 \text{ when } \frac{3}{2} (1 - 2t) = 0$$

$$1 - 2t = 0$$

$$2t = 1$$

$$t = \frac{1}{2}$$

$$\text{Note that } \frac{d^2A}{dt^2} = \frac{3}{2} \times (-2)$$

$$= -3$$

$$\text{i.e. } \frac{d^2A}{dt^2} < 0 \text{ concave DOWN} \curvearrowright$$

 $\therefore$ Area is a MAXIMUM

$$\text{when } t = \frac{1}{2}$$

 $\hookrightarrow$ continued...

Ques 16 cc) (ii) continued:

When $t = \frac{1}{2}$:

$$\begin{aligned} A &= \frac{3}{2} (t - t^2 + 2) \\ &= \frac{3}{2} \left(\frac{1}{2} - \frac{1}{4} + 2 \right) \\ &= \frac{3}{2} \times \frac{9}{4} \\ &= \frac{27}{8} \end{aligned}$$

$$\begin{aligned} \therefore \text{Maximum Area of } \triangle APB \\ &= 3 \frac{3}{8} \text{ square units} \end{aligned}$$

But from question, required to find

$$\begin{aligned} &\frac{3}{4} \times (\text{Area parabolic segment APB}) \\ &= \frac{3}{4} \times 4 \frac{1}{2} \quad (\text{from (i)}) \\ &= \frac{27}{8} \\ &= 3 \frac{3}{8} \text{ square units.} \end{aligned}$$

$$\begin{aligned} \therefore \text{Maximum area of } \triangle APB \\ &= \frac{3}{4} \times (\text{area of parabolic segment APB}) \end{aligned}$$

As required

(4m)